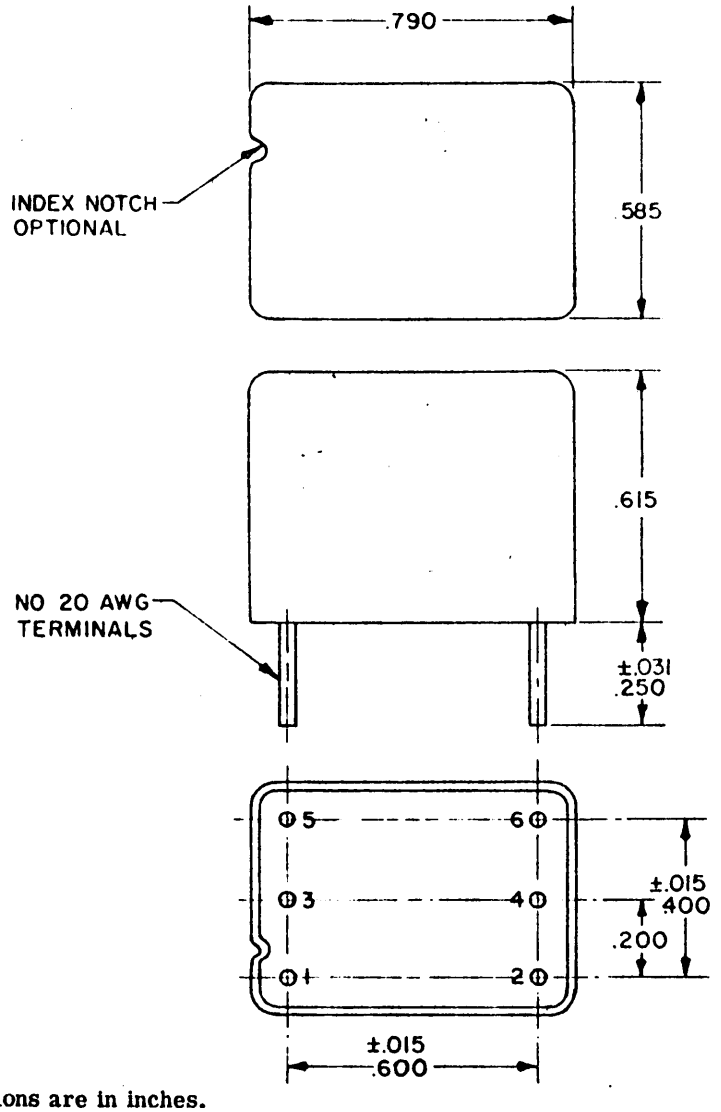
MILITARY SPECIFICATION SHEET

TRANSFORMERS, PULSE, LOW POWER,

TYPE TP7RX2110HZ

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-21038.

Ⓐ This specification is inactive for new design after the date of this specification.



INCHES	MM
.015	.38
.031	.79
.200	5.08
.250	6.35
.400	10.16
.585	14.86
.600	15.24
.615	15.62
.790	20.07

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (.25 mm).

FIGURE 1. Dimensions and configuration.

Ⓐ denotes changes

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TABLE I. Electrical characteristics including waveform parameters.

Dash No.	Impedance (ohms)		Current (mA)		Voltage peak (volts)	DCR (ohms)(max)		Capacitance winding to winding (pF)(max)	Turns ratio	Pulse duration (μs)(min)	Pulse rise time (ns)(max)	Overshoot (% max)	Droop (% max)	Backswing (% max)	Decay time (ns)(max)	Pulse repetition frequency (kHz)	Inductance (primary) (mH)
	(Z_S) Source	(Z_L) Load	Pri	Sec	V_2 Sec	Pri	Sec										
001	200	100	.090	.090	120	.2	.15	24	2:1:1	0.10	40	25	10	15	75	5.0	0.065
002	130	68	.090	.090	45	.35	.2	35	2:1:1	.3	30	25	10	15	50	5.0	0.210
003	130	68	.090	.090	31	1.0	.5	45	2:1:1	.6	40	25	10	15	75	5.0	0.440
004	300	150	.018	.018	31	5.0	2.5	40	2:1:1	1.0	45	25	10	15	100	5.0	2.19
005	1000	200	.036	.036	32	2.75	2.0	45	2:1:1	2.0	100	20	20	15	150	5.0	4.37
006	1000	200	.022	.022	45	10	5.0	130	2:1:1	2.0	180	25	10	15	220	5.0	20.0
007	1600	680	.014	.014	31	22	12.3	150	2:1:1	5.0	160	25	10	15	200	5.0	37.2

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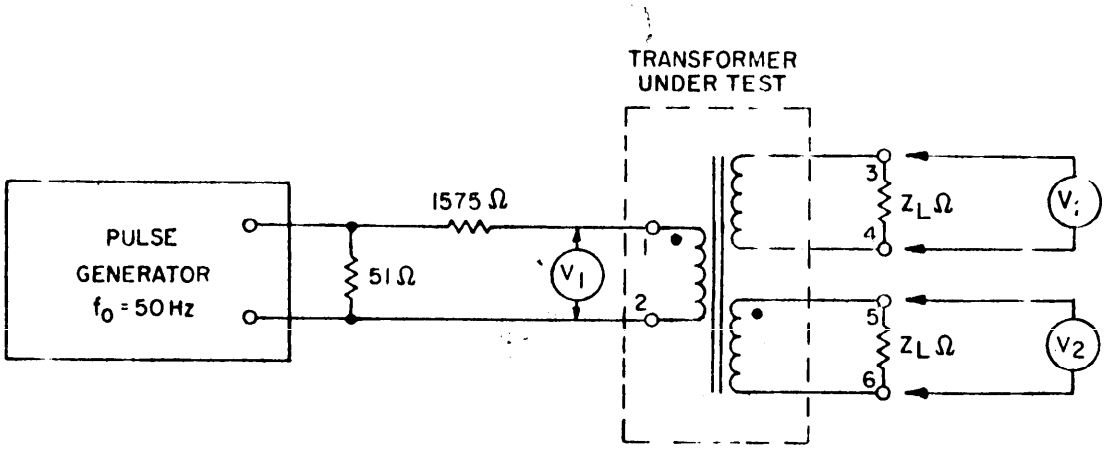


FIGURE 2. Test circuit.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Terminals: Solid-wire leads.

Lead finish: Hot-dip solder per MIL-F-14072, type 1, finish No. M221.

Weight: 12 grams.

Terminal strength: Method 211, MIL-STD-202, test condition A - pull 2.5 pounds.

Dielectric withstanding voltage:

At atmospheric pressure: Method 301, MIL-STD-202, 700 volts rms for 60 seconds between windings and ground.

Barometric pressure (reduced): Method 301, MIL-STD-202, and method 105, test condition F, MIL-STD-202. 100 volts rms between windings and ground.

Altitude: Sea level to 12,000 feet (operating); sea level to 50,000 feet (nonoperating).

Insulation resistance: Method 302, test condition B, MIL-STD-202. Not less than 1,000 megohms.

Electrical characteristics: See table I.

Test circuit: See figure 2.

Life (expectancy "X"): Applicable.

Ambient temperature: Operating, -54°C to 71°C and nonoperating, -62°C to 71°C.

Temperature rise: 34°C.

Resistance to solder wave heat: Method 216, test condition B, MIL-STD-202.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

(A) Quality assurance provisions:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Groups A, B, and C tests of MIL-T-21038 shall be applicable.

Part number: M21038/13-(dash number from table I).

Preparing activity:
Air Force - 11

Agent:
DLA - ES

(Project 5950-F190)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-21038/13A

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